Root Cause Analysis

Here's what the **5 Whys** analysis revealed:

Why #1

Why do critical patients and accident victims deteriorate or die because their ambulance gets trapped in fixed-timer signal queues?

→ Because most traffic signals follow pre-set timing cycles that do not automatically adjust to give priority to approaching ambulances, so emergency vehicles are forced to wait in the same queues as regular traffic.

Why #2

Why don't traffic signals automatically adjust to give priority to approaching ambulances?

→ Because the traffic management system usually has no reliable, real-time way to detect an ambulance's location, direction, and urgency level and translate that into dynamic signal changes at upcoming intersections.

Why #3

Why doesn't the traffic management system have a reliable, real-time way to detect and prioritize ambulances?

→ Because emergency services (ambulance dispatch, hospital systems) and city traffic control operate on separate, legacy platforms with little data integration, no common communication protocol, and no mandated process for sharing priority information about critical patients.

Why #4

Why are emergency services and city traffic control running on separate, non-integrated systems with no mandated priority-sharing process?

→ Because when traffic signal infrastructure and policies were designed and procured, the primary goals were generalized traffic flow and cost-efficiency, not medical emergency responsiveness, so interoperability with health/emergency systems and priority rules for critical patients were never treated as core requirements.

Why #5

Why were emergency medical response needs not treated as core requirements in traffic signal design and policy?

→ Because governance, incentives, and KPIs for traffic systems are typically owned by transport and urban planning bodies that are measured on congestion and throughput, while health outcomes (like survival rates for critical patients) are owned by separate health agencies, leaving no single accountable owner or cross-agency mandate to design traffic control around time-critical medical needs.

Root Cause

Traffic signal systems are governed, specified, and evaluated separately from emergency medical services, so there is no integrated data infrastructure, shared priority protocol, or accountable owner making time-critical medical response a core design requirement of traffic signal control.

RapidRoute Signal Cloud

A cloud-based platform that lets ambulance dispatchers and traffic control centers share live trip data and automatically generate green-wave corridors on city routes. Students can prototype this as a web dashboard where dispatch enters an emergency route and simulated intersections update their signal timings in real time using a standard priority API.

Target User

Ambulance dispatch operators and city traffic control officers in congested Indian cities

Key Feature

A standardized emergency-priority API that pushes time-stamped, route-aware priority requests to all signals on an ambulance's path, turning fragmented controllers into a coordinated corridor.

Impact

How much will this help your target users?



Low impact — High impact

Effort

How many things need to fall in place for this to succeed?



Sirenz Driver Assist App

A mobile app that connects emergency vehicle drivers and nearby commuters to a shared live view of an approaching ambulance and upcoming signal changes. The app guides ambulance drivers along the fastest, signal-aware route while sending commuters in the corridor simple, timed instructions on how to clear space safely as lights preempt for the emergency vehicle.

Target User

Emergency vehicle drivers and daily urban commuters using smartphones

Key Feature

Dual-mode interface where an emergency driver triggers a 'priority corridor' and commuters within a defined radius receive synchronized alerts and lane/stop-line guidance that matches the optimized signal plan.

Impact

How much will this help your target users?



Low impact — High impact

Effort

How many things need to fall in place for this to succeed?

LifeLane Governance Hub

A web-based governance and analytics tool that unifies emergency trip logs, signal performance data, and delay hotspots into a single shared workspace for EMS and traffic departments. It lets officials replay real emergencies on a city map, test new priority rules in a simulator, and assign clear ownership for fixing specific intersections and protocols.

Target User

City traffic planners, emergency medical service administrators, and urban mobility policymakers

Key Feature

A shared, incident-level timeline that links each emergency trip to the exact signals encountered, their states, delays, and responsible agency, making coordination gaps and accountability visible in one integrated dashboard.

Impact

How much will this help your target users?

★ ★ ★ ★ ★

Low impact High impact

Effort

How many things need to fall in place for this to succeed?

★ ★ ★ ★ ★

Compare and select an idea to build

Based on your ratings, here's how the ideas compare

RapidRoute Signal Cloud Recommended

A cloud-based platform that lets ambulance dispatchers and traffic control centers share live trip data and automatically generate green-wave corridors o...

Impact: ★★★★★ Effort: ★★★★

75 score

LifeLane Governance Hub

A web-based governance and analytics tool that unifies emergency trip logs, signal performance data, and delay hotspots into a single shared workspace fo...

Impact: ★★★★ Effort: ★★★

67 score

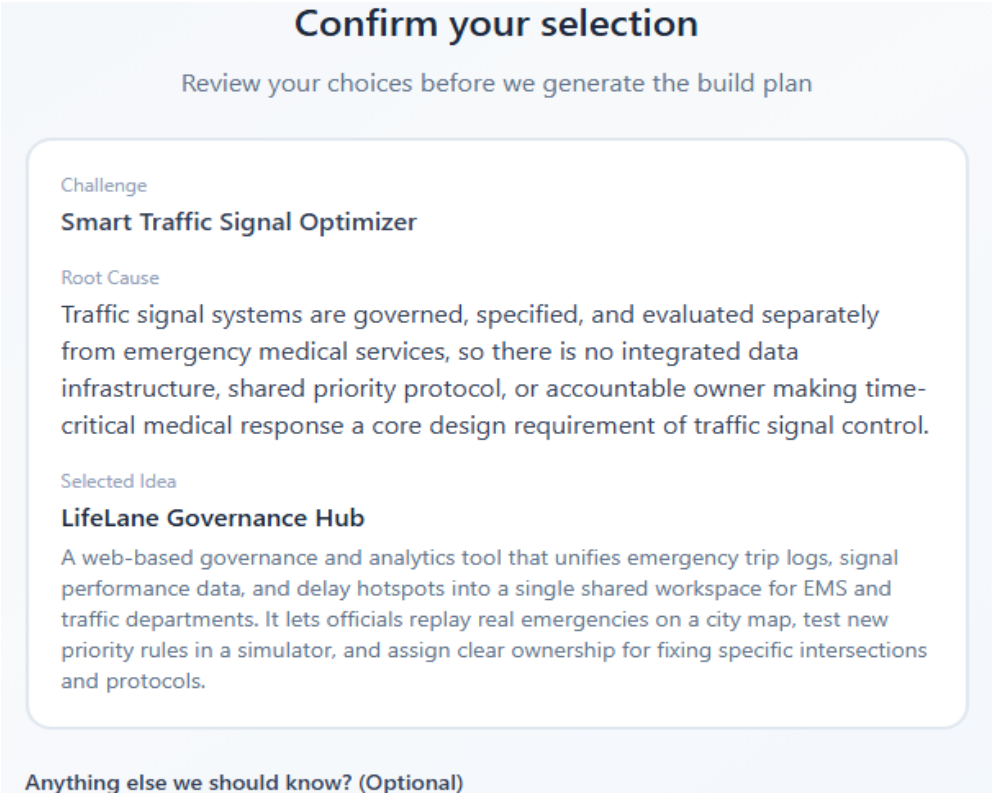
Sirenz Driver Assist App

A mobile app that connects emergency vehicle drivers and nearby commuters to a shared live view of an approaching ambulance and upcoming signal...

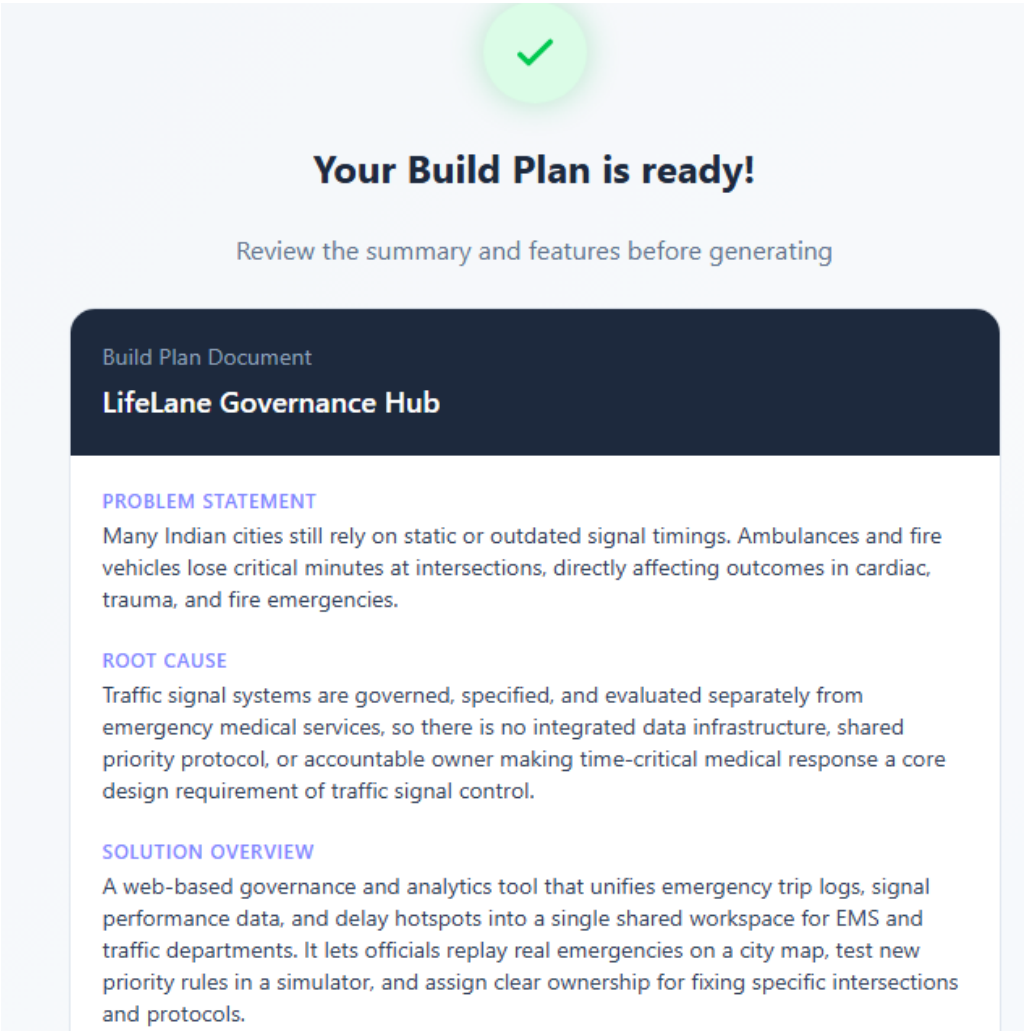
Impact: ★★★★ Effort: ★★★★★

50 score

How we score: Higher impact and lower effort means a better score. The recommended idea offers the best balance.



For the build phase, the prototype will be constrained to a dashboard-style data visualization using generated historical datasets. To ensure a polished, 10/10 deliverable within the hackathon timeframe, the focus will be on building the analytics interface (visualizing delay hotspots and EMS trip logs via tools like Power BI, Tableau, or Python) rather than developing a live, real-time mapping web application from scratch. The core deliverable will be demonstrating the data logic and governance metrics.



TARGET USERS

City traffic planners, emergency medical service administrators, and urban mobility policymakers

FEATURES TO BUILD

- **Incident-Level Timeline Dashboard**
  A comprehensive dashboard that provides a unified timeline of emergency trips, linking each event to the corresponding traffic signals encountered. This feature highlights signal states, delays, and the responsible agency to improve coordination and accountability.

- **Emergency Scenario Replay**
  An interactive map-based feature allowing users to replay past emergency scenarios. Users can visualize the route taken, traffic signal states at each intersection, and identify delay hotspots, aiding in analysis and decision-making.

- **Signal Performance Analytics**
  A tool to analyze traffic signal performance during emergencies, providing metrics such as average delay times and signal responsiveness. This feature enables stakeholders to identify patterns and prioritize improvements.

- **Priority Rule Simulator**
  A simulation tool to test new traffic signal priority rules. Users can simulate potential changes to signal protocols and visualize their impact on emergency response times in a virtual environment.

- **Ownership and Accountability Tracker**
  A feature that assigns and displays ownership of specific intersections and protocol improvements, allowing users to track accountability for traffic signal optimization efforts and ensure appropriate follow-up actions.